Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel International Award in Lower Secondary

Tuesday 28 May 2024

Afternoon (Time: 1 hour 20 minutes)

Paper reference **LCP11/01**

Computing

Year 9
Achievement Test

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- You must **NOT** use a calculator.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 State **two** ways of staying safe **when using** social media.

1 .....

.....

2 .....

.....

(Total for Question 1 = 2 marks)

2 Data can be sent between computers using networks.

(a) Complete the sentence using the correct option from this list.

(1)

- filter software
- search engines
- web browsers
- wireless protocols

The use of ..... can stop users seeing unwanted online content.

(b) Which **one** of these is included in the header of a data packet?

(1)

☒ **A** GPS location data

☒ **B** HTML

☒ **C** Sequence number

☒ **D** Server URL

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(c) State **one** advantage of using a wireless rather than a wired network.

(1)

.....

.....

(Total for Question 2 = 3 marks)

3 State **one** way that technology can be used to track where someone goes.

.....

(Total for Question 3 = 1 mark)

4 Which **one** of these is a protocol used to create a personal area network?

☒ **A** Bluetooth

☒ **B** NFC

☒ **C** Satellite

☒ **D** WAP

(Total for Question 4 = 1 mark)

5 Complete the sentence using the correct option from this list.

- data speed
- latency
- security
- signal strength

Mbps is the unit used to measure the ..... of a network.

(Total for Question 5 = 1 mark)

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6 Complete the sentence using the correct option from this list.

- GPS
- HTTPS
- NFC
- Wi-Fi

A reason for using ..... for payments is that it has a range of 4 centimetres.

(Total for Question 6 = 1 mark)

7 Network devices are used to connect computers.

(a) State the purpose of a modem.

(1)

.....

.....

(b) State **one** way people use networks to collaborate.

(1)

.....

.....

(Total for Question 7 = 2 marks)

8 State the role of an input device in a computer system.

.....

(Total for Question 8 = 1 mark)

9 Which **one** of these is a type of software licence that allows users to modify the code?

☒ **A** Malware

☒ **B** Open source

☒ **C** Proprietary

☒ **D** Utility

(Total for Question 9 = 1 mark)

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10 Computers use hardware and software.

(a) Complete the sentence using the correct words from this list.

- application
- interface
- parameter
- user
- utility

(2)

An operating system provides the ..... between the computer and the ..... .

(b) Complete the sentence using the correct option from this list.

- operating system
- optical disc drive
- read-only memory
- secondary storage

(1)

The ..... stores a computer's boot sequence.

(c) Which **one** of these executes instructions?

(1)

☒ **A** Central processing unit

☒ **B** Hard-disk drive

☒ **C** Output device

☒ **D** Random-access memory

(d) State **one** type of anti-malware.

(1)

.....

(Total for Question 10 = 5 marks)

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11 Draw **one** straight line from **each** storage device to its correct use.

Storage device	Use
Hard-disk drive	Online file storage
	Outputting data
Optical disc drive	Storing files to read-only media
	Internal storage for a smartphone

(Total for Question 11 = 2 marks)

12 Computers use binary to represent information.

(a) Which **one** of these is the number of kibibytes in one mebibyte?

(1)

- ☒ **A** 0.5
- ☒ **B** 10
- ☒ **C** 1000
- ☒ **D** 1024

(b) Convert the denary integer 87 to 8-bit binary.

(2)

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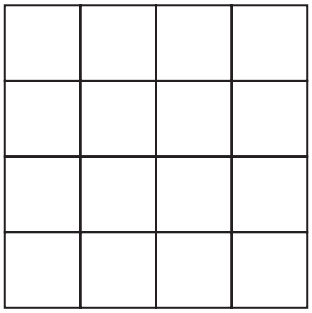
DO NOT WRITE IN THIS AREA

(c) An image is represented by this binary pattern.

0110 1101 1011 1001

Complete the grid to show the image represented by the binary pattern.

(2)



(Total for Question 12 = 5 marks)

13 Data representation is often not perfect.

(a) State **two** ways to improve the binary representation of sound.

(2)

1 .....

.....

2 .....

.....

(b) Describe how colour is represented in a bitmap image.

(2)

.....

.....

.....

.....

(Total for Question 13 = 4 marks)

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14 The ASCII code for the character B is 66.

Give the character represented by the ASCII code 70.

.....

(Total for Question 14 = 1 mark)

15 The Python programming language can make use of lists.

Here is some Python code.

```
1 points = [5, 7, 3, 12]
2 answer = points[0] * 2
3 print (answer)
```

(a) Which **one** of these is used in this code?

(1)

- ☒ **A** Iteration
- ☒ **B** Repetition
- ☒ **C** Sequence
- ☒ **D** Selection

(b) State the name of **one** variable used in this code.

(1)

.....

(c) State the purpose of the arithmetic operator used in this code.

(1)

.....

(d) Which **one** of these does `points[0]` refer to?

(1)

- ☒ **A** All values
- ☒ **B** First value
- ☒ **C** Last value
- ☒ **D** No value

(Total for Question 15 = 4 marks)

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16 Here is some Python code.

```
3 print (animal.lower()
```

(a) Complete the sentence using the correct word from this list.

- binary
- logical
- sorting
- syntax

(1)

The missing closing bracket causes a ..... error.

(b) Which **one** of these is the type of data that is altered with `.lower()`?

(1)

☒ **A** ASCII

☒ **B** bit

☒ **C** string

☒ **D** subprogram

(Total for Question 16 = 2 marks)

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17 Here is some Python code.

```
1  number = input ("Enter a number: ")
2  n = int (number)
3  for i in range (1, n+1):
4     print (i)
```

(a) State the data type returned by the input function.

(1)

.....

(b) Give the output of this code after a user enters the number 3.

(2)

.....

.....

.....

.....

(Total for Question 17 = 3 marks)

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18 Complete the output for each line of code in the table.

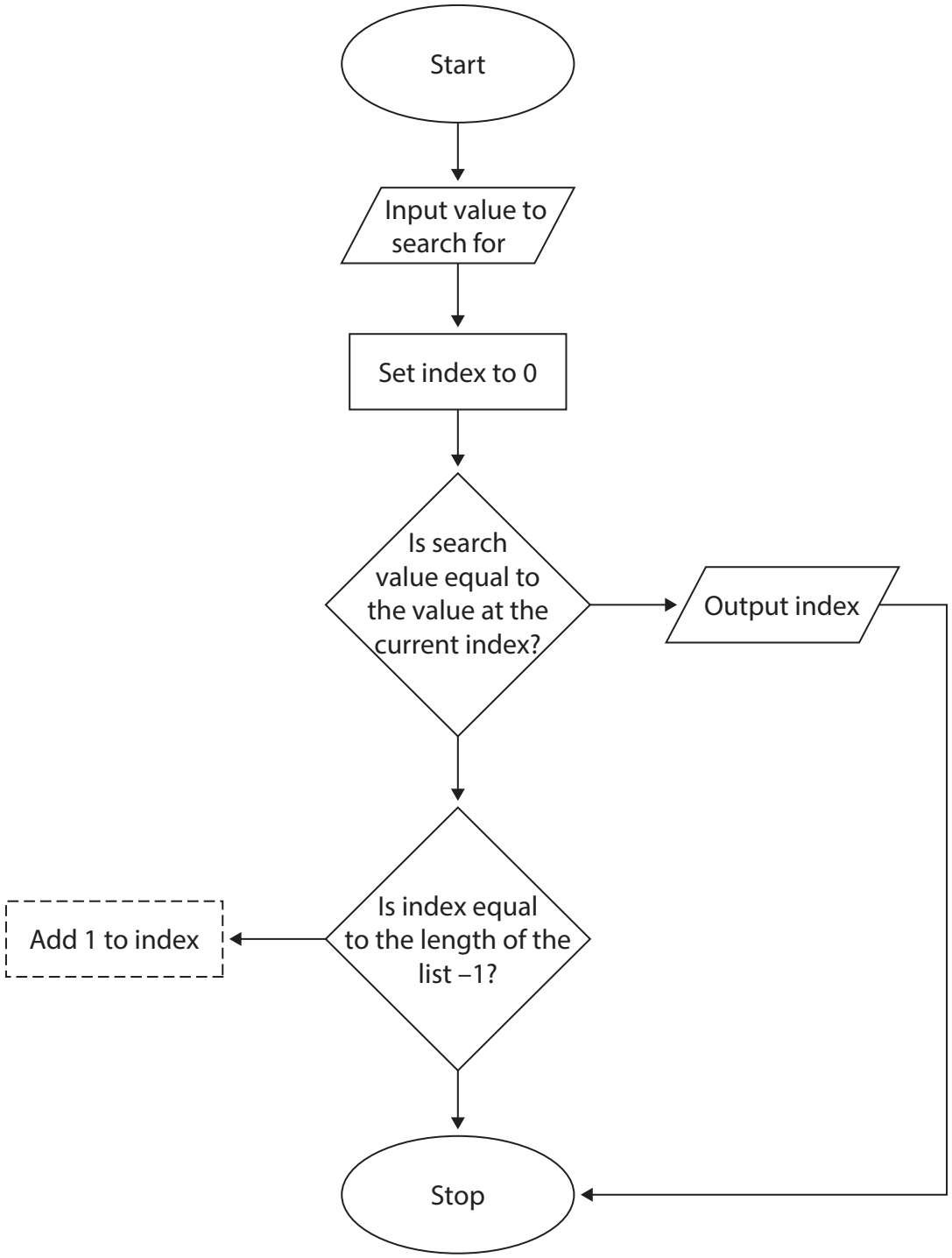
The first one has been done for you.

Code	Output
`print (0 > 2)`	`False`
`print (0 == 2)`	
`print (0 != "")`	

(Total for Question 18 = 2 marks)

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19 Here is an incomplete flowchart, which represents a linear search algorithm that looks for a value in a list.



(a) Complete the flowchart by adding:

- Yes and No labels on the lines from each decision box
- an arrow line from the process box with a dashed outline to create a loop.

(3)

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(b) Explain why the second decision is needed.

(2)

.....

.....

.....

.....

(Total for Question 19 = 5 marks)

20 Here is some Python code.

The code represents a guessing game.

Complete the program by adding Python code on the dotted lines.

```
1  number = 5
2  guess = 0
3  ..... guess != number:
4     guess = ..... ("Guess a number from 1 to 10: ")
5     guess = int(.....)
6     if guess > number:
7        print ("Lower")
8     elif guess ..... number:
9        print ("Higher")
10 print ("Correct!")
```

(Total for Question 20 = 4 marks)

TOTAL FOR SECTION A = 50 MARKS

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

21 Buttons are often included in presentations.

(a) Give **one other** type of content used in presentations.

(1)

.....

(b) Complete the sentence using the correct option from this list.

- alignment
- comments
- speaker notes
- transitions

(1)

Presentations use ..... when moving between slides.

(Total for Question 21 = 2 marks)

22 A company uses word processing to create documents.

(a) Complete the sentences using the correct words from this list.

- letter
- memo
- newsletter
- poster
- spreadsheet

(2)

A ..... is a multi-page, word-processed document.

It may use a column layout to provide information.

A ..... is a short document used to present information to a team within a company.

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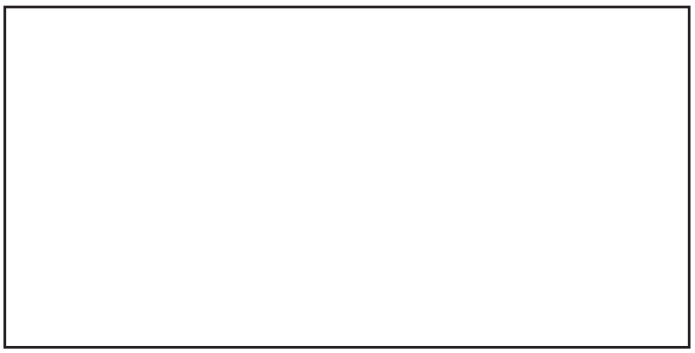
(b) State **one** type of text alignment.

(1)

.....

(Total for Question 22 = 3 marks)

23 Here is a text box.



(a) Circle **one** part of the text box that can be used to resize it.

(1)

(b) State **one** benefit of using a text box to include text in a document.

(1)

.....

.....

(Total for Question 23 = 2 marks)

24 Complete the sentence using the correct option from this list.

- colour range
- compression ratio
- file size
- pixel count

One benefit of a vector graphics file is the .....

is smaller.

(Total for Question 24 = 1 mark)

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25 Which **one** of these can happen when enlarging a bitmap file?

- ☒ **A** Cropping
- ☒ **B** Enhancement
- ☒ **C** Erasing
- ☒ **D** Pixelation

(Total for Question 25 = 1 mark)

26 A web page needs a bitmap image. It must download quickly.

Explain the most appropriate choice of file type for the image.

.....

.....

.....

.....

(Total for Question 26 = 2 marks)

27 Databases can be flat file or relational.

(a) Which **one** of these is **only** used in a **relational** database?

(1)

- ☒ **A** Foreign key
- ☒ **B** Query
- ☒ **C** Record
- ☒ **D** Relational operator

(b) State **one other** difference between a flat file database and a relational database.

(1)

.....

.....

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(c) Which **one** of these is an example of a **logical** data value stored in a database?

(1)

- ☒ **A** 5
- ☒ **B** t
- ☒ **C** ÷
- ☒ **D** Yes

(d) A database contains this information about books:

- title
- author
- genre
- publication_year

Which **one** of these is the single criterion you would use to search for **all** books published in 2014?

(1)

- ☒ **A** 2014 is year
- ☒ **B** publication_year = 2014
- ☒ **C** title <> 2014 and publication_year = 2014
- ☒ **D** title <> 2014, author <> 2014, genre <> 2014

(Total for Question 27 = 4 marks)

28 Which **one** of these is a reason for using a file compression tool?

- ☒ **A** To convert files from one format to another
- ☒ **B** To create backups so data can be restored
- ☒ **C** To encrypt files to protect them from unauthorised access
- ☒ **D** To make files smaller so they use less storage

(Total for Question 28 = 1 mark)

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29 You share a file with a friend but do not want the file to be edited.

State the file permission you should give your friend.

.....

(Total for Question 29 = 1 mark)

30 Web pages use HTML.

(a) Which **one** of these is used in a web page to create a location that can be linked to?

(1)

- ☒ **A** Bookmark
- ☒ **B** Button
- ☒ **C** Hyperlink
- ☒ **D** Thumbnail

(b) Describe the purpose of a <head> tag on a web page.

(2)

.....

.....

.....

.....

(Total for Question 30 = 3 marks)

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31 Look at **Figure 1**.

It shows part of a spreadsheet used to track swimming records.

	A	B	C
1	**Swimmer**	**50m freestyle**	**Date achieved**
2	Alex	29.65	June 2, 2022
3	Maya	30.12	May 10, 2021
4	Liam	30.48	March 5, 2023
5	Emma	31.02	July 15, 2022
6	Noah	31.55	April 22, 2023
7	Average	30.56	
8	Fastest	29.65	
9	Slowest		

Figure 1

(a) Which **one** of these is the formatting type used in cell B3?

(1)

- ☒ **A** Currency
- ☒ **B** Decimal
- ☒ **C** Date/Time
- ☒ **D** Text

(b) State **one** reason why a graph cannot be created from the data in column A.

(1)

.....

(c) The formula =SUM(B2:B6)/5 is used in cell B7 to give the average time for the 50m freestyle.

Explain why this formula should **not** be used if other swimmers' times are added.

(2)

.....

.....

.....

.....

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(d) Complete the function that should be used in cell B8 to show the fastest time.

(2)

= ( )

(e) Which **one** of these is the use of a chart legend?

(1)

- ☒ **A** To add value labels
- ☒ **B** To display a meaningful title
- ☒ **C** To label the x and y axes
- ☒ **D** To colour-code categories of data

(Total for Question 31 = 7 marks)

32 A company needs an animated advertisement.

(a) State the best choice of software application to create this.

(1)

.....

(b) Explain why the company should use a house style for all documents.

(2)

.....

.....

.....

.....

(Total for Question 32 = 3 marks)

TOTAL FOR SECTION B = 30 MARKS
TOTAL FOR PAPER = 80 MARKS

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